

EARS, HEARING, AND BALANCE

THE EARS PROVIDE THE SENSE OF HEARING. THEY ALSO DETECT HEAD POSITION AND MOTION, SO ARE ESSENTIAL TO BALANCE. THE FUNCTION OF BOTH HEARING AND BALANCE IS BASED ON THE ACTIVITY OF “HAIR CELL” RECEPTORS.

INSIDE THE EAR

The ear is divided into three parts. The outer ear comprises the ear flap and the slightly S-shaped outer ear canal, which guides sound waves to the second region, the middle ear. The elements of the middle ear amplify the sound waves and transfer them from the air into the fluid of the inner ear. They include the eardrum and the three smallest bones in the body: the auditory ossicles, which span the

air-filled cavity of the middle ear. The fluid-filled inner ear changes sound waves to nerve signals inside the snail-shaped cochlea. The middle-ear cavity connects to the throat via the Eustachian tube, and so to the air outside. This connection allows atmospheric pressure to transfer to the cavity, equalizing the air pressure on either side of the eardrum and preventing it from bulging as the outside pressure changes.

